

Master Product Requirements Document (PRD): OptumAuth-AI

Doc Ref: PRD-OPTUM-2026-V3	Owning Dept: Optum Product Management	Sprint Cycle: Release Train 2026.3	Status: READY FOR SPRINT EXECUTION
-------------------------------	--	---------------------------------------	---------------------------------------

1. User Personas & Target Journey

Primary Persona: Utilization Management Nurse (Optum/UMR). Reviews 60–80 incoming clinical prior authorization requests daily across Epic and Optum Provider Portal. Pain point: 78% of time spent scrolling 50-page PDF charts searching for lab and therapy trial dates.

2. Functional Requirements (FR) Traceability Matrix

REQ ID	FUNCTIONAL REQUIREMENT SCOPE	PRIORITY	ACCEPTANCE GATE
FR-1	Automated FHIR R4 Bundle Ingestion via Da Vinci CRD/DTR CDS Hooks.	P0	Payload ingested and parsed in ≤ 1.5s
FR-2	Multi-Tenant Summary Plan Description (SPD) Vector Resolution for UMR groups.	P0	100% custom employer exclusions flagged
FR-3	Deterministic LangGraph extraction with sentence-level bounding-box provenance.	P0	JSON schema adherence = 100%, Temp = 0.0
FR-4	SMART-on-FHIR Split-Screen Verification UI with interactive yellow highlighting.	P0	One-click jumping from criterion to PDF page
FR-5	Direct ASC X12N 278 EDI transaction generation and Cirrus/Facets injection.	P0	Real-time auto-adjudication status posted

3. Agile User Story & Gherkin Scenario

Feature: Clinical Evidence Extraction & Citation Highlighting

Scenario: Extraction of Completed Physical Therapy
Given a de-identified FHIR bundle containing note: "Pt completed 8 weeks of PT at ActiveMotion on 11/02/2025"
When extraction agent runs against InterQual Criterion C ("Conservative Management >= 6 weeks")
Then criterion status must return "MET" with confidence >= 0.90
And return exact character offsets: "char_start": 820, "char_end": 935
And render interactive yellow bounding box over source sentence in SMART-on-FHIR panel.

4. Product Acceptance & Sign-Off

ROLE	NAME	SIGN-OFF STATUS	DATE
Lead AI Product Manager	Abhinav Kumar	PRD BASELINED	2026-03-26
Engineering Director	Vikram Malhotra	CAPACITY COMMITTED	2026-03-27
Lead Clinical Designer	Sarah Jenkins	UX PROTOTYPE VALIDATED	2026-03-26